

Introduction to Non Offload Advertisement and Scan feature

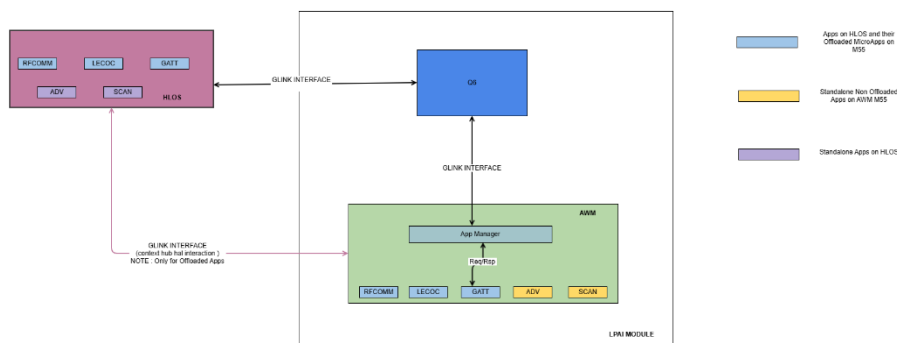
This section explains the high level and lower level details for Advertisement and Scanning Apps on AWM which interacts with BT Host on ADSP to perform scanning and advertisement operations on LPAI. The feature request allows for scan and advertisement apps to independently register and perform operations from AWM with any interventions from HLOS side . This is different from other apps like RFCOMM and LECOC (offloaded apps) which are also present on AWM but are tied with an parent app on HLOS in order to perform their operations.

Non Offloaded App

With the current BT architecture on Aspen, we have an concept of offloaded apps . These apps are the ones where the primary app resides on the HLOS and offloads some of its functionalities on the offloaded apps on low power islands (apps on AWM). For these offloaded apps , the primary control with respect to connections resides with the HLOS and only operations like Tx and Rx are performed by apps on low power island.

However , we also have a concept of non offloaded apps on AWM(Ambient Wear Microcontroller). These apps are only present on AWM and do not have a corresponding entity on HLOS . Thus , the complete control of these apps resides within the AWM subsystem . These apps can be used once BT is enabled and AWM receives that information from ADSP over Glink. The apps can then be used to perform operations on AWM.

Block Diagram



All interactions for Non Offloaded apps will be handled only via Glink interface between AWM and Q6 and do not involve any interaction with HLOS.

Interactions for Offloaded apps are initiated once offload is initiated from HLOS.

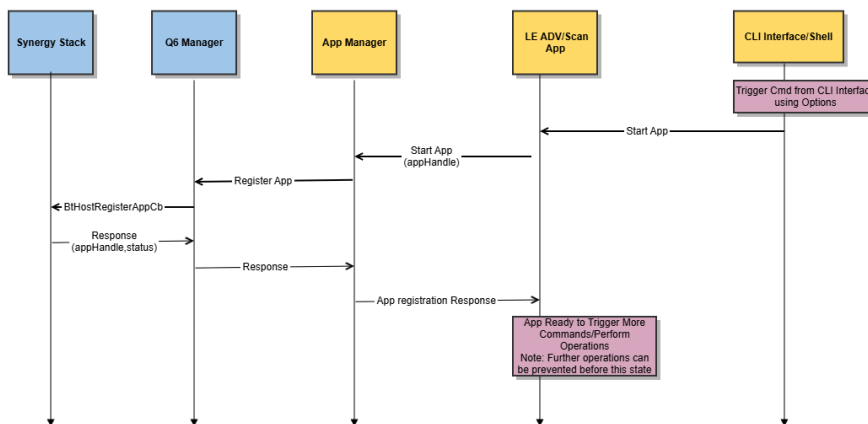
Note : For Scan and Advertisements , the parameters set by Android HLOS will take precedence over parameters set by microstack.

Functionality Supported

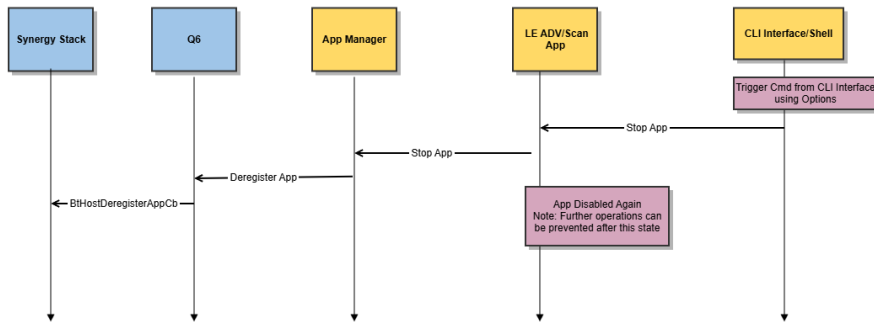
#	Functionality	Description
1.	Register App on M55 and Q6 for app activation	This allows for LE Scan and Advertisement Apps on AWM to register with App Manager on AWM and subsequent enablement on ADSP for performing operations
2.	UnRegister App on M55 and Q6 for app deactivation	This allows for LE Scan and Advertisement Apps on AWM to un-register with App Manager on AWM and subsequent disablement on ADSP.
3.	Enable Scanning	This allows for client on AWM to start scanning for devices and receive advertisement reports
4.	Disable Scanning	This allows for client on AWM to stop scanning for devices
5.	Enable Advertisement	This allows client app on AWM to start advertisement with specific params and data.
6.	Disable Advertisement	This allows client app on AWM to stop advertisement.

Message Sequences

Start Non_Offload App



Stop Non_Offload App



LE Advertisement

For Le Advertisement App on microapp , the following features will be provided.

Number of Advertisements Sets supported will be limited to 2

Only Extended Advertisements will be supported . Legacy and Periodic Advertisements will not be supported.

Maximum Advertisement Data Length Supported : 251 Bytes

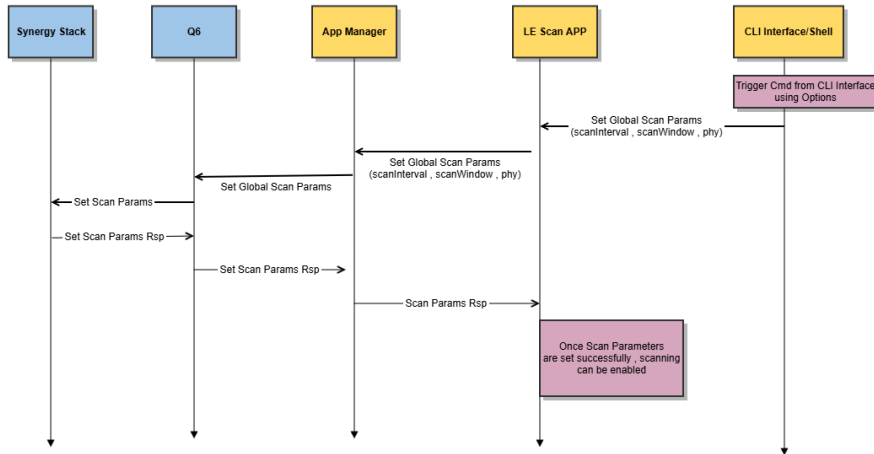
Note: Connectable Advertisements will not be supported

- Register an LE ADV SET
- Set Parameters for an ADV SET
- Set Advertisement Data for an ADV SET
- Enable an ADV SET
- Disable an ADV SET

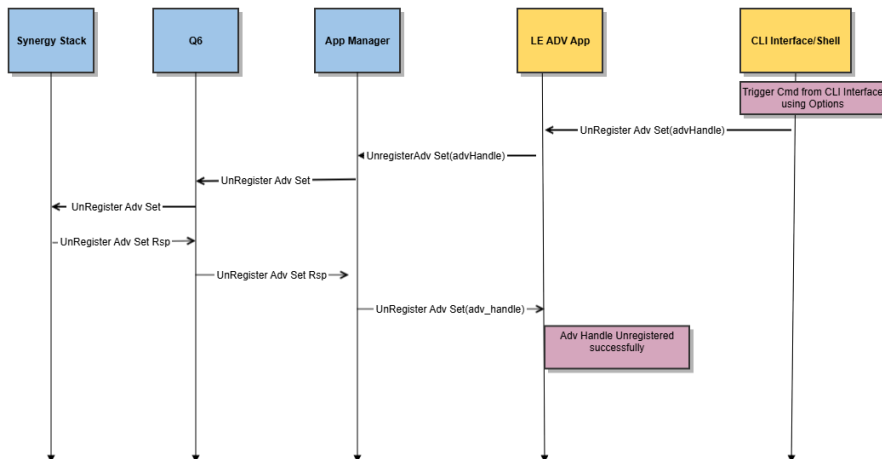
#	Functionality	Description
1.	Register Le Adv Set	This allows for registration of an Advertisement Set on ADSP
2.	Set Parameters for Advertisement Set	This allows setting advertisement parameters for an advertisement set
3.	Set Data for an Advertisement Set	This allows setting advertisement data for an advertisement set
4.	Enable Advertisement Set	This allows for client on AWM to enable an advertisement set
5.	Disable Advertisement Set	This allows for client on AWM to disable an advertisement set

Message Sequences for BLE Advertisements

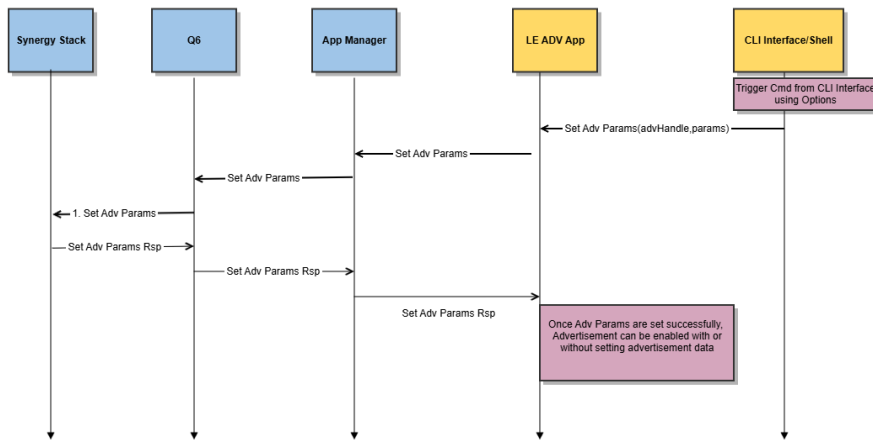
Register Adv Set



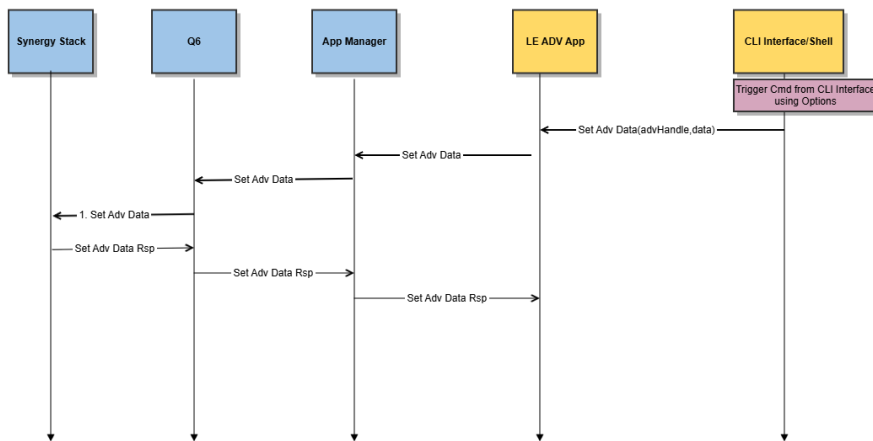
Unregister Adv Set



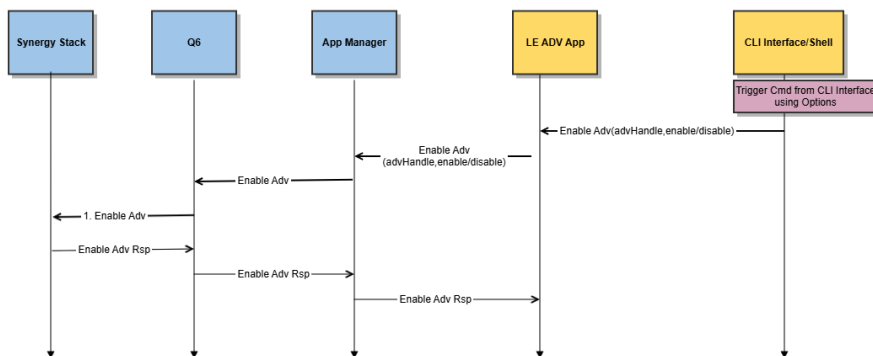
Set Advertisement Params



Set Advertisement Data



Enable/Disable Advertisement Set



LE Scanning

For Le Scanning App on microapp , the following features will be provided.

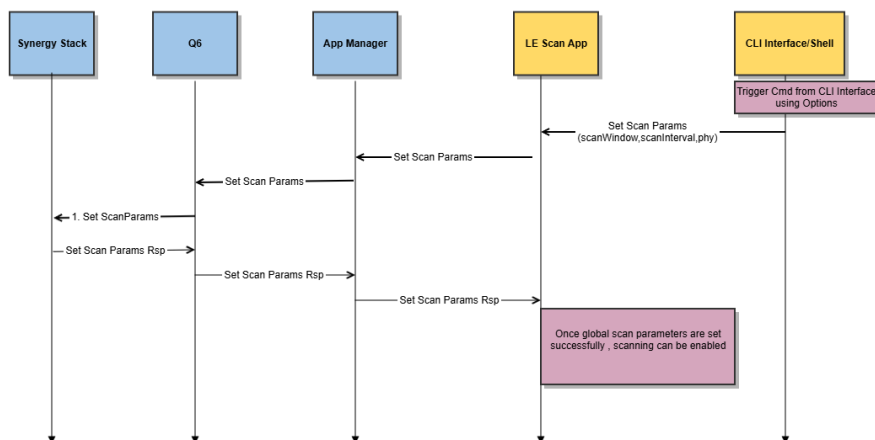
- Register an LE Scanner
- Set Global Scan Parameters
- Enable Scanner
- Disable Scanner

#	Functionality	Description
1.	Register Le Scanner	This allows for registration of an Advertisement Set on ADSP
2.	Set Global Scan Parameters	This allows setting advertisement parameters for an advertisement set
3.	Enable Scanner	This allows to enable scanning
4.	Disable Scanner	This allows to disable scanning

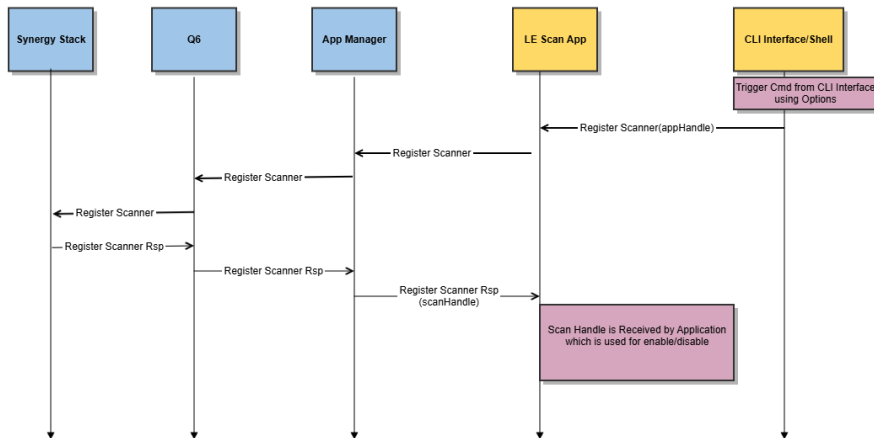
Note: Any filtering based on specific parameters(uuid , local name) is not provided as it is not supported by the Host.

Message Sequences

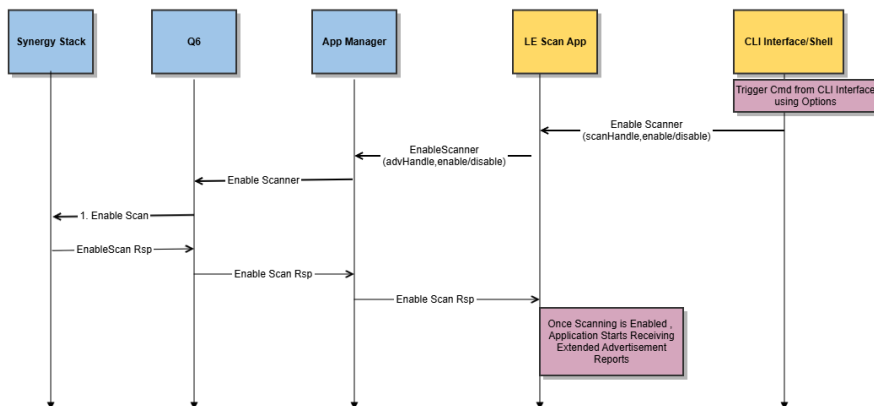
Set Global Scan Parameters



Register Scanner



Enable/Disable Scanner



Interface details Between ADSP and AWM for exchanging Messages

```

/**
 * @brief Structure to stop an advertisement.
 * @param appHandle Application handle
 * @param advHandle Advertising handle
 */
typedef struct Stop_Advertisement{
    uint16_t appHandle;
    uint8_t advHandle;
}Stop_Advertisement_t;
    
```

```

/**
 * @brief Structure to get App Handle for ADV App Registration with BT HOST
 * @param appHandle Application Handle
 * @param status Bool to Specify if App Registration was successful
 * */
typedef struct startAdvAppRsp{
    uint16_t appHandle;
    bool status;
}startAdvAppRsp_t;

/**
 * @brief Structure to confirm the registration of an advertising set.
 * @param resultCode Result code for registration of an advertising set
 * @param advHandle Advertising handle
 * */
typedef struct{
    uint16_t resultCode;
    uint8_t advHandle;
}RegisterAdvSetCfm;

/**
 * @brief Structure to confirm the unregistration of an advertising set.
 * @param resultCode Result code for unregistration of an advertising set
 * @param advHandle Advertising handle
 * */
typedef struct{
    uint16_t resultCode;
    uint8_t advHandle;
}UnregisterAdvSetCfm;

/**
 * @brief Structure to confirm the start of an advertisement.
 * @param resultCode Result code of the start process
 * @param advBits advBits
 * @param maxAdvSets Maximum number of advertising sets
 * */
typedef struct Start_Adv_Cfm{
    uint16_t resultCode;
    uint32_t advBits;
    uint8_t maxAdvSets;
}Start_Adv_Cfm_t;

/**
 * @brief Structure to confirm the stop of an advertisement.
 * @param appHandle Application handle
 * @param status Status of the stop process
 * */
typedef struct Stop_Adv_Cfm{
    uint16_t appHandle;
    uint16_t status;
}Stop_Adv_Cfm_t;

```

```

/**
 * @brief Structure to register a scan set.
 * @param flags Flags for scan set registration.
 * @param scan_filter Scan filter
 */
typedef struct registerScanner{
uint16_t adv_filter;
}registerScanner_t;

/**
 * @brief Structure to unregister a scan set.
 * @param scanHandle Scan handle.
 */
typedef struct unregisterScanner{
uint16_t scanHandle;
}unregisterScanner_t;

/**
 * @brief Structure to set scan parameters.
 * @param appHandle Application handle.
 * @param flags Flags for scan parameters.
 * @param ownAddrType Own address type.
 */
typedef struct setGlobalScanParams{
uint8_t ownAddrType;
uint8_t scanFilterPolicy;
uint8_t scanningPhys;
uint8_t filterDuplicates;
uint8_t scanType;
uint16_t scanInterval;
uint16_t scanWindow;
}setGlobalScanParams_t;

typedef struct enableScan{
bool enable;
uint8_t scanHandle;
}enableScan_t;

/**
 * @brief Structure to enable scanning.
 * @param scanHandle Scan handle.
 * @param scanTime Duration of the scan.
 * @param appHandle Application handle.
 */
typedef struct setGlobalParamsCfm{
uint16_t type;
uint16_t resultCode;
}setGlobalParamsCfm_t;

typedef struct registerScannerCfm{
uint16_t type;
uint16_t resultCode;
uint8_t scanHandle;
}registerScannerCfm_t;

```

```
typedef struct unregisterScannerCfm{
uint16_t type;
uint16_t resultCode;
}unregisterScannerCfm_t;
```

```
typedef struct enableScanCfm{
uint16_t type;
uint16_t resultCode;
}enableScanCfm_t;
```

QAPI's details:

```
/**
 *@brief This Api Allows for Registration of Advertisement App
 */
void qapi_registerLeAdvApp();

/**
 *@brief This Api Allows for Registration of Advertisement Set which can be
 used for further operations
 */
void qapi_registerAdvSet();

/**
 *@brief This Api Allows for Registration of Advertisement Set which can be
 used for further operations
 *@param[in] advHandle Advertisement Handle
 *@param[in] advEventProperties Advertisement Type
 *@param[in] advIntervalMin Minimum Adv Interval
 *@param[in] advIntervalMax Maximum Adv Interval
 *@param[in] advTxPower Tx Power
 */
void qapi_setAdvParams(uint8_t advHandle , uint16_t advEventProperties ,
uint16_t advIntervalMin , uint16_t advIntervalMax , uint16_t advTxPower);

/**
 *@brief This Api Allows for Registration of Advertisement Set which can be
 used for further operations
 *@param[in] advHandle Advertisement Handle
 *@param[in] advData Data to be Set
 */
void qapi_setAdvData(uint8_t advHandle , uint8_t *advData);

/**
 *@brief This Api Allows for Registration of Advertisement Set which can be
 used for further operations
 *@param[in] advHandle Advertisement Handle
 *@param[in] enable bool to enable Disable Adv Set
 */
void qapi_enableAdvSet(uint8_t advHandle , bool enable);
```

```

/**
 * @brief This Api Allows for Registration of Scanner App
 */
void qapi_registerLeScanApp()

/**
 * @brief This Api Allows for Registration of Advertisement Set which can be
        used for further operations
 */
void qapi_registerScanner()

/**
 * @brief This Api is used to set the Global Scan Parameters
 * @param[in] scanFilterPolicy    Scanning Filter Policy to be Used
 * @param[in] scanningPhys        Scanning Phys
 * @param[in] filterDuplicates    To specify if duplicates are needed or not
 * @param[in] scanType            Scan Type : Passive or Active
 * @param[in] scanInterval        Scan Interval Duration
 * @param[in] scanWindow          Scan Window Duration
 */
void qapi_setGlobalScanParams(uint8_t scanFilterPolicy , uint8_t
scanningPhys , uint8_t filterDuplicates , uint8_t scanType , uint16_t
scanInterval , uint16_t scanWindow);

/**
 * @brief This Api is used to enable /disable scan
 * @param[in] scanHandle Scan Handle for which operation is to be performed
 * @param[in] enable      bool to Specify if scanning is to be enabled/disabled
 */
void qapi_enableScan(uint8_t scanHandle , bool enable)

```

Validation of BLE Advertisements and Scanning offload to Micro Synergy Stack

Precondition: BT is Enabled in Offload Mode and Information is relayed to AWM via Glink

Commands to Validate the Advertisements

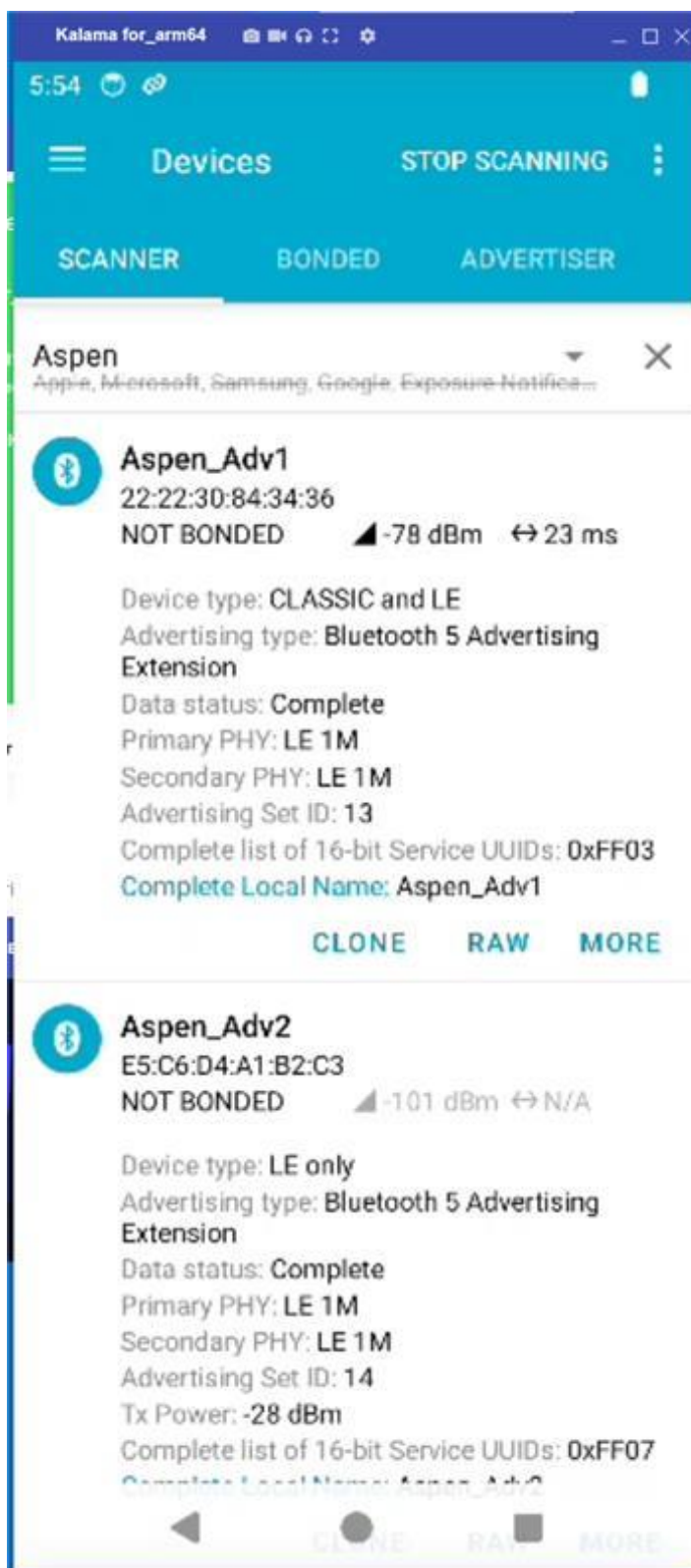
#	Command	Command Usage on Shell
1	Start Adv App	bluetooth startAdvApp

2	Register Adv Set	bluetooth regAdvSet
3	Set Adv Params	Adv Params : <Adv Handle> <advEventProperties <Valid Values : 0 and 64>> <minInterval <Valid Values : 0x20 to 0xFFFFFFFF> > <maxInterval <Valid Values : 0x20 to 0xFFFFFFFF> > <advTxPower : -127 to +20 > <ownAddrType <Valid < 0 : Public , 1: Random >> Ex Usage : bluetooth setAdvParams 15 0 0x20 0x20 0 0 <pre> uart:~\$ uart:~\$ uart:~\$ bluetooth setAdvParams <Adv Handle> <advEventProperties <Valid Values : 0 and 64>> <minInterval <Valid Values : 0x20 to 0xFFFFFFFF> > <maxInterval <Valid Values : 0x20 to 0xFFFFFFFF> > <a dvTxPower : -127 to +20 > <ownAddrType <Valid < 0 : Public , 1: Random >> uart:~\$ bluetooth setAdvParams 15 0 0x20 0x20 0 0 Micoapp Glink Tx : 0 Handling Events for NONOFFLOADED APPS Event Id : 20492 QAPI SET ADV PARAMS CFM Adv Handle : 15 , Result Code : 0 , Tx Power : -1 uart:~\$ uart:~\$ uart:~\$ uart:~\$ uart:~\$ </pre>
4	Set Random Address	Adv Random Address bluetooth setAdvRandomAddr 16 0 0xa1b2c3 0xd4 0xe5c6 <pre> uart:~\$ bluetooth setAdvRandomAddr <Adv Handle> <addrType <Valid : Static : 0> > <lap <Ex:0xa1b > > <nap <Ex:0xe5f6>> uart:~\$ bluetooth setAdvRandomAddr 16 0 0xa1b2c3 0xd4 0xe5c6 Micoapp Glink Tx : 0 Handling Events for NONOFFLOADED APPS Event Id : 20494 QAPI SET RANDOM ADDRESS CFM Adv Handle : 16 , Result Code : 0 , Lap : 2632c , Uap : 7b , uart:~\$ uart:~\$ </pre>

5	Set Adv Data	Adv Data <AdvHandle> <Service_uuid <Ex:0xff01> > <Local Name <Max Len : 20> > Ex Usage : bluetooth setAdvData 15 0xff03 Aspen Adv1 <pre> uart:~\$ uart:~\$ uart:~\$ bluetooth setAdvData <AdvHandle> <Service_uuid <Ex:0xff01> > <Local Name <Max Len : 20> > uart:~\$ uart:~\$ bluetooth setAdvData 0xff03 Aspen Adv1 <AdvHandle> <Service_uuid <Ex:0xff01> > <Local Name <Max Len : 20> > uart:~\$ uart:~\$ bluetooth setAdvData 15 0xff03 Aspen Adv1 Size of Local Name : 10 Size of Local Name : 10 Microapp Glink Tx : 0 Handling Events for MONOFFLOADED APPS Event Id : 20493 QAPI SET ADV DATA CFM Adv Handle : 15 , Result Code : 0 uart:~\$ uart:~\$ </pre>
6	Enable Advertisement	bluetooth enableAdv <Adv_Handle> Ex: bluetooth enableAdv 16
7	Disable Advertisement	bluetooth disableAdv <Adv_Handle> Ex: bluetooth disableAdv 16

Result

One advertiser enabled with Public Address and other advertiser enabled with random address



Limitations

- Send only UUID and name during setting of advertisement data due to Glink Limitation between Q6 and AWM
- **Note:** Change will be made once Glink between Q6 and AWM is stable

Scan Commands Usage

#	Command	Command Usage on Shell
1	Start Scan App	bluetooth startScanApp
2	Register Scanner	bluetooth regScanner Usage : <adv_filter : Valid Values : 0 to 3> Adv Filter Values - 0 - Receive all advertising data. Uses sub field below. 1 - Do not receive any advertising data. 2 - Only receive legacy advertising data (Primary advertising channel). Uses sub field below 3 - Only receive extended advertising data (Secondary advertising channel). Ex Usage : bluetooth regScanner 0
3	Enable/Disable Scanner	Enable Scanner : <Enable/Disable> <Scan_Handle> <duration> Duration Values : 0 - Scanning until disabled 0x01 - 0xFFFF - RFU Ex Usage : bluetooth enableScan 1 1 0
4	Unregister Scanner	bluetooth regScanner <scanHandle> Ex Usage : bluetooth unregScanner 1

Limitations

- Receive Extended Scan Reports with limited parameters due to Glink Limitation between Q6 and AWM
- **Note:** Change will be made once Glink between Q6 and AWM is stable